

In the electron diffraction experiment, electrons are emitted via thermionic emission from a current carrying wire and are incident on a crystal, for example graphite. The aim is to determine the atomic spacing by analysing the circles of constructive interference seen on a phosphor screen due to the diffracted electrons.

If the electrons are accelerated through a potential V the conservation of energy gives:

$$eV = \frac{1}{2}mv^2$$

De-Broglie's formula for the wavelength of a 'matter wave' used to model the wave-like behavior of particles is:

$$p = \frac{h}{\lambda}$$

Thus since $p = mv$ by definition the electron's 'wavelength' is:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Bragg's law relates this wavelength to the atomic spacing d . If waves are incident at angle θ to atoms in a grid along the horizontal the path difference is $2d \sin \theta$ and the condition for constructive interference is:

$$2d \sin \theta = n\lambda$$

where n is any integer and is the order of diffraction. Since the electrons are incident along the horizontal in the electron diffraction experiment, they will be at an angle $\phi = 2\theta$ when diffracted towards the rings of constructive interference, an angle that can easily be measured by measuring $\tan \phi$ using a vernier caliper or on an image. Thus the atomic spacing can be calculated from:

$$d = \frac{nh}{2\sqrt{2meV} \sin \frac{\phi}{2}}$$